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Do professionals' inflation forecasts incorporate the beliefs of others? A functional data approach☆

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ABSTRACT

We apply functional data analysis to survey expectations data, and show that functional principal component analysis combined with functional regression analysis is a fruitful way of capturing the effects of others' forecasts on a respondent's inflation forecasts. We estimate forward-looking Phillips curves on each respondent's inflation and unemployment rate forecasts, and show that for nearly half of the respondents, the forecasts of others are important. The functional principal components of the cross-sectional distributions of forecasts are shown to capture characteristics other than the mean or consensus forecast, and include forecaster disagreement.

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1. Introduction

In the recent literature, there has been much interest in why survey respondents disagree with one another in terms of their expectations.[1] Our interest is in whether forecasters disagree because they have different views (or models) about how the economy operates. We consider inflation forecasting, and whether professional forecasters operate with different Phillips curve models in mind, in terms of the responsiveness of inflation to the degree of 'slack' in the economy. We focus on the Phillips curve, given its central role in academic research and macroeconomic modeling since Phillips (1958) first drew attention to the inverse relationship between UK wage inflation and unemployment.[2] The novelty of our approach, relative to other studies of the extent to which professional forecasters profess a belief in a Phillips curve,[3] is that we simultaneously allow for the possibility that professional forecasters are aware of, and may take into account, the forecasts of others when they form their expectations.

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[1] See Giacomini et al. (2020) for an analysis of professional forecasters allowing for different channels of heterogeneity, including heterogeneous priors, heterogeneous inattentiveness, and the use of different models.

[2] Other papers consider the theory-consistency of agents' expectations with a relationship between relative consumption growth and real exchange-rate depreciations across countries (see Backus and Smith (1993)), or whether consumers' spending decisions are consistent with an Euler equation (see Dräger and Nghiem (2021), *inter alia*).

[3] See e.g. Casey (2020) and Clements (2024), *inter alia*, who fit Phillips curve-type models to the forecasts of survey respondents at the individual level.

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Failure to allow for this possibility may provide a misleading picture of the extent to which forecasts conform to Phillips curve models.

We use functional data analysis (FDA) to include the forecasts of others in individual forecasters' Phillips curve models. Our rationale for using FDA is similar to that of Meeks and Monti (2023). Meeks and Monti (2023) use functional data analysis to capture the dynamic heterogeneity of beliefs and how these affect inflation dynamics. Functional factors are used in place of the consensus expectation commonly used in New Keynesian Phillips curve models, arguing that 'the average expectation may not adequately summarize the state of beliefs relevant to the determination of inflation, leading to errors of mismeasurement in econometric models' (p. 42).

We do not know how individual forecasters are affected by the forecasts of others, and falsely assuming that only the consensus view matters, for example, may result in errors of mismeasurement in our setting. To avoid any potential information loss, in principle one would need the entire distribution of all the other individuals' forecasts. However, conventional econometric models cannot deal with data objects in the form of continuous curves, such as probability density functions or cumulative distribution functions. FDA does allow for the use of objects that can be naturally viewed as curves and is a feasible method for extracting information from the cross-sectional distribution of forecasts, and for investigating whether professional forecasters' Phillips curves incorporate such information.

Since we do not know at the outset whether simply including the consensus forecast (say) contains all relevant information, we believe that a flexible approach is warranted which allows the data to 'speak'. FDA is one such approach. In this paper we show how FDA can be applied in our context, and we compare FDA with more traditional approaches, such as including a selection of moments of the cross-sectional distribution.

Looking ahead, we find that professional forecasters do respond to the forecasts of others. However, it transpires that whether an allowance is made for the forecasts of others has little effect on our estimates of respondents' perceptions of the responsiveness of inflation to the unemployment rate. Our FDA approach proves to be successful in capturing the effects of others' forecasts. As it turns out, the overall picture would have been similar had we instead used a selection of cross-sectional moments in place of the functional factors, although there are some differences, as we discuss. However, the two approaches need not have given similar results, and we believe FDA to be a better-founded approach in general. We believe that further empirical studies might help to establish when FDA would be expected to outperform the use of cross-sectional moments.

Our baseline model is a relatively simple constant-parameter linear relationship between inflation and the unemployment rate, although the Phillips curve was originally proposed by Phillips (1958) as an inverse relationship between UK wage inflation and unemployment. Following Calvo (1983) it has emerged as a micro-founded relationship between inflation, expectations of future inflation, and a measure of slack: the so-called expectations-augmented Phillips curve.[4] The simplicity of our baseline model reflects the relatively small number of forecasts typically available at the level of the individual forecaster. There is evidence that the actual Phillips curve (between actual inflation and a measure of slack in the economy, as opposed to individuals' forecasts of these quantities) may have flattened over time, and that the relationship may be non-linear (see e.g. Hooper et al. (2019)). Indeed, for forecasts, Fendel et al. (2011) find that the slope of the Phillips curve depends on the state of the business cycle. We consider extensions to the basic model that allow for possible dependence on the phase of the business cycle, and other non-linearities, and find strong empirical support for doing so.

To make matters concrete, the baseline Phillips curve model estimated for each respondent j is (following Clements (2024)):

$$E_{j,t}\pi_{t+h} = \beta_j E_{j,t}\pi_{t+h+1} + \alpha_j E_{j,t}u_{t+h} + e_{j,t}, \quad (1)$$

where $E_{j,t}\pi_{t+h}$ is j's h-step ahead forecast of π_{t+h} made at time t, where π is the quarterly inflation rate, and $E_{j,t}u_{t+h}$ is the forecast of the unemployment rate u. Thus, (1) is a relationship that would be expected to hold (subject to a disturbance term $e_{j,t}$) between individual j's forecasts of inflation (for periods $t+h$ and $t+h+1$) and the unemployment rate, all made at time t, if j's forecasts are consistent with a Phillips curve-type relationship.

In addition to the range of Phillips curve models that might guide expectations, there is a large literature suggesting that forecasters pay attention to the views of others for a variety of reasons (see e.g. Clements (2018)). This is usually allowed for in the macro-forecasting literature by including the consensus or average of the (appropriately dated) cross-section of individuals' forecasts. One of the aims of our paper is to consider whether alternative ways of including others' forecasts might give rise to different findings, or whether findings using consensus forecasts are reasonably robust.

The proposal is to extend (1) to allow for the forecasts of others in a flexible way. That is, we allow j's forecasts to be driven by an aggregator function of other individuals' forecasts (made at time $t-1$, so that these are known to j at time t). Because we do not know how professional forecasters use the information contained in the forecasts of others, we proceed by including the information in the entire distribution of all the individuals' time $t-1$ forecasts. Thus, in addition to including $E_{j,t}\pi_{t+h+1}$ on the right-hand side (as suggested by the expectations-augmented Phillips curve applied to an individual's forecasts), we also include $\int \gamma_j(x)dF_{t+h|t-1}(x)$ in (1), which gives us:

$$E_{j,t}\pi_{t+h} = \beta_j E_{j,t}\pi_{t+h+1} + \int \gamma_j(x)dF_{t+h|t-1}(x) + \alpha_j E_{j,t}u_{t+h} + e_{j,t}, \quad (2)$$

where $F_{t+h|t-1}$ is the distribution of individuals' forecasts made at time $t-1$ for time $t+h$, and $\gamma_j(\cdot)$ is the coefficient function which serves as the flexible aggregator for the individual j. Notice that $\gamma_j(\cdot)$ is subscripted by j. This allows

[4] See Gali and Gertler (1999) and Coibion et al. (2018) for a review of its historical development.

individual *j*'s forecasts to be influenced by the forecasts of others in a way which is peculiar to that individual.

The challenges to standard methods of model estimation and inference in (2) resulting from the inclusion of the distribution of individuals' forecasts can be overcome by the use of FDA. FDA deals with problems where one of the variables in the analysis can be naturally viewed as a smooth curve or function (Horváth & Kokoszka, 2012; Ramsay & Silverman, 2005). That is, FDA can be thought of as the statistical analysis of samples of curves, possibly combined with the quantities of standard analysis: vectors and scalars. FDA has witnessed rapid development over the last two decades. While the central ideas and methods of FDA have achieved a certain maturity, applications of FDA proceed at an accelerating speed. Economics and finance have benefited from the use of tools from FDA. Some of the prominent examples in economics and finance include: intraday price curves (Kokoszka et al., 2015), the term structure of interest rates (Bardsley et al., 2017; Horváth et al., 2022), forward curves of commodity futures (Horváth et al., 2020), and price signatures (Oomen, 2019). In our case, the distribution of individuals' forecasts can be characterized as a function. We show how the tools developed in FDA can be applied to make statistical inference on (2).

As mentioned above, we follow Meeks and Monti (2023) in using FDA to extract information from the cross-section of forecasts. However, they relate distributions of inflation expectations from surveys of forecasters to actual inflation. They apply functional principal component analysis (FPCA) to the distributions of inflation expectations and find that the first three principal components (PCs) can be interpreted as disagreement, skew, and shape. They then estimate Phillips curve models of actual inflation with forward-looking components comprising these PCs, as a way of taking into account the individual-level heterogeneity. Their approach is founded within the literature that uses survey expectations as a source of expectations external to the model itself, obviating the need to instrument future values of variables.[5] Our study instead seeks to understand how professional survey respondents' forecasts of inflation are affected by the forecasts of others. To the best of our knowledge, it is the first to apply FDA to the analysis of the theory-consistency of macro-expectations. FDA is employed to capture potential interactions between forecasters.[6]

Our interest is in professional forecasters. Coibion and Gorodnichenko (2015) argue that using firms' expectations in the Phillips curve instead of those of professional forecasters better explains the so-called missing disinflation following the 2008 financial crisis, but our interest is in understanding the formation of the expectations themselves. We suppose that professional forecasters are more likely to be cognizant of modern macro-theory than consumers, for example, and will be more likely to monitor sources of information which may include the views of fellow professionals.

Finally, we acknowledge that our findings that forecasters look at the forecasts of others are consistent with explanations of forecaster behavior, such as that of Ilut and Schneider (2014) (and see also Masolo and Monti (2021)), who suppose forecasters entertain multiple priors, and that shocks to confidence, measured by the dispersion of forecasts, reflect ambiguity (Knightian uncertainty).[7]

The plan of the rest of the paper is as follows. Section 2 describes the forecast data we use, and how and why the empirical cross-sectional inflation forecast distributions are transformed to cross-sectional log quantile densities (LQDs) of the inflation forecasts. Section 3 explains how we apply FDA to estimations of the Phillips curve models of the form of (2), for each respondent *j*, and discusses the findings. Section 4 then explores the robustness of our findings to some of the modeling choices we made, and to alternative formulations. Section 5 offers some concluding remarks, as well as some potential avenues for future research. We believe the methods we discuss and illustrate might be fruitfully used in many other contexts relating to survey expectations.

2. Descriptive analysis

The forecast data are from the US Survey of Professional Forecasters (SPF), the longest-running survey of macro-expectations of professionals.[8] The survey began in 1968:Q4, and has been held quarterly ever since. The individual-level data are made available on the website of the Federal Reserve Bank of Philadelphia. Extensive use of the SPF data has been made by academic researchers. For our purposes, it is important that the respondent to a given survey will know the responses made to the survey in the previous quarter, as well as the fact that individuals can be tracked through time even though they remain anonymous. We use the individual expectations from 1981:Q3 to 2022:Q2, inclusive, for:

- CPI inflation rate (CPI) headline: annualized percentage points, seasonally adjusted, based on quarterly average index level.
- Civilian unemployment rate (UNEMP): percentage points, seasonally adjusted, quarterly average.

The cross-sectional distributions use the forecasts from all the respondents, some of whom make only a small number of returns, although we only estimate Phillips curve models for respondents who make more than a minimum number of forecasts.

2.1. Properties of the forecasts

Table 1 presents the summary statistics of the SPF CPI and unemployment forecasts. Individuals give their forecasts of the current quarter (i.e. the quarter of the

[5] See for example McCallum (1976) and Coibion et al. (2018).

[6] See Clements (2024) for a discussion of the literature on investigating whether macro-forecasts are consistent with various macro-theories, including the Phillips curve.

[7] We are grateful to a reviewer for making this point.

[8] Clements et al. (2022) provide a review of surveys of professional forecasters. On the U.S. SPF, see Croushore (1993) and Croushore and Stark (2001).

Table 1
Summary statistics of CPI and unemployment forecasts.

	CPI								
	Mean	SD	Skewness	Kurtosis	10% Q	25% Q	50% Q	75% Q	90% Q
$h=0$	2.839	0.907	−0.202	3.875	1.822	2.346	2.855	3.388	3.848
$h=1$	2.866	0.802	−0.306	4.492	2.014	2.488	2.893	3.298	3.725
$h=2$	2.931	0.748	−0.223	4.480	2.156	2.596	2.943	3.324	3.723
$h=3$	2.992	0.744	−0.072	4.582	2.218	2.639	2.998	3.367	3.797
$h=4$	3.047	0.754	−0.011	4.509	2.252	2.685	3.051	3.422	3.859
	Unemployment								
	Mean	SD	Skewness	Kurtosis	10% Q	25% Q	50% Q	75% Q	90% Q
$h=0$	6.217	0.142	0.211	3.925	6.064	6.142	6.212	6.294	6.376
$h=1$	6.180	0.226	0.088	3.388	5.924	6.051	6.175	6.297	6.446
$h=2$	6.125	0.287	0.066	3.470	5.801	5.947	6.123	6.283	6.463
$h=3$	6.067	0.338	0.084	3.549	5.690	5.865	6.062	6.254	6.464
$h=4$	6.009	0.383	0.112	3.599	5.585	5.781	6.000	6.225	6.456

survey, denoted $h=0$), of the next quarter ($h=1$), and of the next three quarters. The table pools the forecasts across individuals and survey quarters. It is evident that the standard deviation of the CPI forecasts is greatest at $h=0$, whereas the reverse is true for the unemployment rate forecasts.

2.2. Data transformation

We begin by plotting the cross-sectional distributions of the inflation forecasts for each quarter in the sample (1981:Q3 to 2022:Q2) for $h=1$, where the density functions have been estimated using a kernel-type estimator; see the upper panel of Fig. 1.

However, density functions satisfy two constraints: (1) strict non-negativity and (2) integrating to one, which complicate their use in FDA.[9] The literature on FDA has devoted much effort to how best to deal with density curves. An influential paper is Petersen and Müller (2016). Their main idea is to suitably transform the densities (constrained curves) into unconstrained curves, so that one can embed the densities into an L^2 space. Then, standard methods of FDA, such as FPCA and functional regression (detailed below), can be applied to the transformed curves in the L^2 space. If all the densities share a common known support $\mathscr{T}$, the representations of the density curves themselves can be obtained by applying the inverse transform to the unconstrained curves in the L^2 space. Petersen and Müller (2016) recommend two choices for the functional transformation: the log hazard transform, and the log quantile density (LQD) transformation. The LQD transformation has become the more popular choice. It has been used in e.g. bioscience (Petersen et al., 2019) and engineering (Chen et al., 2019), and we use it in our application.

Briefly, the LQD transformation works as follows. A cross-sectional distribution at time t is commonly represented in one of three ways: a cumulative distribution function (CDF) F_t, a probability density function (PDF) f_t, or a quantile function Q_t. The three representations have the following relationship:

$$F_t(x) = \int_{-\infty}^{x} f_t(u)du, \quad -\infty < x < \infty,$$
$$Q_t(s) = F_t^{-1}(s) = \inf\{y \in (0,1) : F_t(y) \geq s\}, \quad s \in (0,1).$$

A given distribution is uniquely characterized by any of the three functions F_t, f_t, and Q_t, each of which may be used as a functional data object. Nevertheless, all three are subject to nonlinear constraints. Specifically, the CDF is constrained by $0 \leq F_t(x) \leq 1$ and $F_t'(x) \geq 0$; the PDF by the constraints $f_t > 0$ and $\int_{-\infty}^{\infty} f_t(u)du = 1$; while the quantile function is the least constrained, requiring only that $Q_t'(x) \geq 0$. Unfortunately, the presence of the nonlinear constraints complicates the application of typical linear functional data methods, as noted above. To enable the application of FDA, we consider ways to characterize a distribution with fewer or no constraints. Jones (1992) proposes the so-called quantile density, $q_t = Q_t'$, which is only constrained to be nonnegative. The definition of the quantile density implies:

$$q_t(s) = Q_t'(s) = (F_t^{-1})'(s) = \frac{1}{F_t'(F_t^{-1}(s))} = \frac{1}{f_t(Q_t(s))}, \quad s \in (0,1).$$

Then, an unconstrained representation for a distribution is obtained by taking the log of the quantile density: this gives the LQD

$$X_t(s) = \log(q_t(s)) = -\log(f_t(Q_t(s))), \quad s \in (0,1). \tag{3}$$

One drawback of Petersen and Müller (2016) is that all densities are assumed to have the same support, to enable the LQD in the L^2 space to be mapped back to the density space. However, this assumption is sometimes too restrictive, and fails to hold in some applications. In a follow-up study, Kokoszka et al. (2019) proposed a modified LQD transformation that allows for densities with different supports to be transformed to functions on a common domain. The only requirement for their modified LQD transformation is that the intersection of all the densities' supports contains a known point.

In our study, the densities of professional survey respondents' forecasts of inflation at different times t do not have the same support: the minimum and maximum of the inflation forecasts are not the same across quarters. Nevertheless, we can use the Petersen and Müller

[9] Density functions can be viewed as elements of a Hilbert space, but they do not constitute a *linear* subspace. If the densities are treated as elements of an L^2 space, then linear combinations of densities will be elements of L^2, but will not necessarily be densities. Thus, it is not appropriate to directly apply conventional FDA approaches to density curves.

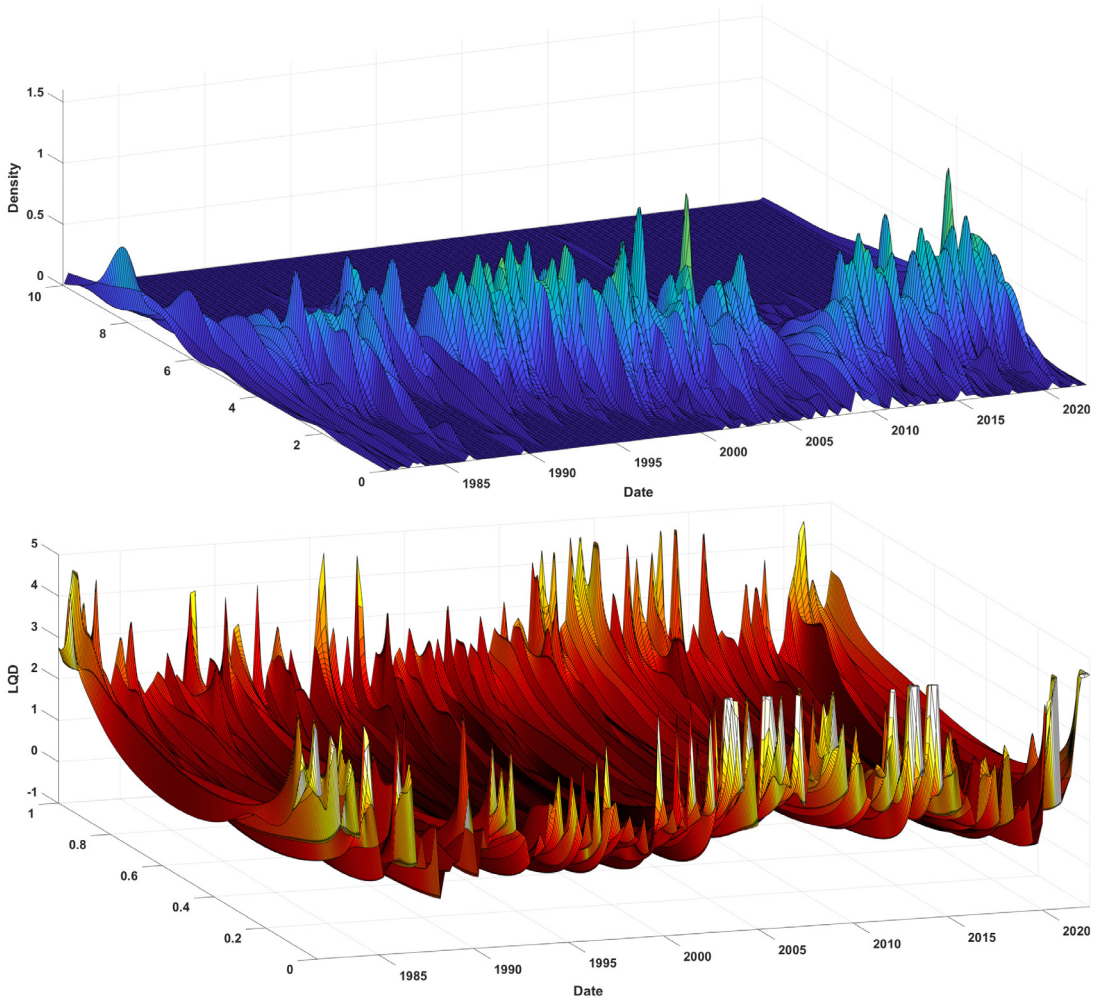


Fig. 1. Upper panel: Cross-sectional densities of inflation forecasts for $h = 1$. Lower panel: Cross-sectional LQD of inflation forecasts for $h = 1$.

(2016) version of the LQD transformation rather than the modified version. This is because we are interested in how the distribution of individuals' forecasts may affect a particular respondent, and we do not need to invert LQDs back to the density space. Thus, the inversion step of Petersen and Müller (2016) is not needed in our study.[10]

Estimating the LQD requires the specification of a kernel function and a bandwidth. In this analysis, we employ a Gaussian kernel and utilize Silverman's (1986) rule of thumb for bandwidth selection.[11] Furthermore, we estimate the LQD using raw cross-sectional inflation forecasts rather than mean-centered data. Another consideration is whether or not to remove outliers. Technically, it becomes necessary to remove outliers when working with the LQD. This is because LQD can go to infinity if any $s \in (0, 1)$ gives $f_t(Q_t(s)) = 0$. Keeping extreme outliers can bring this troublesome issue to the fore. For this reason, outliers at the 1% level in each quarter t for a given h are removed.[12]

In order to evaluate the stationarity of the series of LQDs, we applied the functional KPSS test developed by Horváth et al. (2014) separately for $h = 0, 1, 2, 3, 4$. The p-value of the null hypothesis of stationarity is as follows: 0.241 ($h = 0$), 0.451 ($h = 1$), 0.417 ($h = 2$), 0.258 ($h = 3$), and 0.140 ($h = 4$). The results suggest that it is reasonable to assume that the series of LQDs are stationary, which provides the justification for the functional regression model.

A visualization of the cross-sectional LQD of the inflation forecasts at each quarter of our sample period for $h = 1$ is provided in the lower panel of Fig. 1. The LQD is typically 'U'-shaped due to the negative sign in its construction (3). Most of the LQD curves are relatively smooth, but some occasionally have small wiggles. Importantly, we can observe that the LQD is always in the domain of $s \in (0, 1)$, which facilitates the functional data

[10] The Kokoszka et al. (2019) modified LQD transformation would not be applicable in any case because the intersection of all densities' supports does not contain a known point.

[11] Using other choices of kernel functions and bandwidth does not affect our main conclusions.

[12] We provide a robustness check of removing outliers at 5% in Appendix F.

analysis. We stress that we work with the LQD, rather than the densities, in the rest of the study.

3. Functional data analysis

In this section we describe our application of FDA to the SPF data. There are two steps. The first is the use of FPCA to capture the main patterns in the variability of the distribution of professionals' forecasts. The second step uses functional linear regression to determine whether an individual's forecasts are affected by the forecasts of the other respondents, and which aspects of the distributions of the forecasts of the other respondents are relevant. One might suppose that the 'consensus' view would be important—that is, the mean of the distribution of the forecasts of the other respondents. Although we find a role for the mean, FDA suggests that the findings are more nuanced.

We begin with the basic model. The different forecast horizons reported in the SPF allow Model (2) to be estimated for left-hand-side forecasts of $h = 0, 1, 2, 3$.[13] Model (2) can be written in full for each h as:

$$\begin{aligned}
E_{j,t}\pi_t &= \zeta_{j,0} + \beta_{j,0}E_{j,t}\pi_{t+1} + \int \gamma_{j,0}(s)X_{t|t-1}(s)ds \\
&\quad + \alpha_{j,0}E_{j,t}u_t + e_{j,t,0}, \\
E_{j,t}\pi_{t+1} &= \zeta_{j,1} + \beta_{j,1}E_{j,t}\pi_{t+2} + \int \gamma_{j,1}(s)X_{t+1|t-1}(s)ds \\
&\quad + \alpha_{j,1}E_{j,t}u_{t+1} + e_{j,t,1}, \\
E_{j,t}\pi_{t+2} &= \zeta_{j,2} + \beta_{j,2}E_{j,t}\pi_{t+3} + \int \gamma_{j,2}(s)X_{t+2|t-1}(s)ds \\
&\quad + \alpha_{j,2}E_{j,t}u_{t+2} + e_{j,t,2}, \\
E_{j,t}\pi_{t+3} &= \zeta_{j,3} + \beta_{j,3}E_{j,t}\pi_{t+4} + \int \gamma_{j,3}(s)X_{t+3|t-1}(s)ds \\
&\quad + \alpha_{j,3}E_{j,t}u_{t+3} + e_{j,t,3}, \quad s \in (0, 1),
\end{aligned} \tag{4}$$

where constant terms have been added, and $X_{t+h|t-1}$ is the LQD of the individual forecasts made at time $t-1$ of the value of inflation at time $t+h$. As written, the model parameters vary across respondent j and horizon h. Following Jain (2019) and Clements (2024), we assume the parameters are the same across h for a given respondent, giving rise to more precise parameter estimates. Doing so gives the pooled version of Model (2), written as:

$$\begin{aligned}
E_{j,t}\pi_{t+h} &= \zeta_j + \beta_j E_{j,t}\pi_{t+h+1} + \int \gamma_j(s)X_{t+h|t-1}(s)ds \\
&\quad + \alpha_j E_{j,t}u_{t+h} + e_{j,t}, \text{ for } h = 0, 1, 2, 3.
\end{aligned} \tag{5}$$

In Appendix C, we show the results of allowing for horizon fixed effects. Generally, doing so makes little difference, justifying their omission in what follows.

[13] Technically, $h = -1$ could also be estimated. However, there is little cross-sectional variation in $E_{j,t}\pi_{t-1}$. This is because the advance estimate of the previous period has already been officially released, and is used by most respondents. Thus we do not include $h = -1$ in the analysis. Note that $h = 0$ means a forecast for the current quarter, which is typically termed 'nowcasting'.

We could further restrict the heterogeneity in the parameters by assuming that the slope parameters are the same across respondents in either (4) or (5) (and possibly using a fixed-effects panel data estimator). We prefer not to impose slope homogeneity across j, so do not use a panel estimator. Both Jain (2019) and Clements (2024) report considerable heterogeneity across j: Jain (2019) reports heterogeneity in respondents' beliefs regarding inflation persistence, and Clements (2024) considers the extent to which heterogeneity in individuals' beliefs in the Phillips curve reflects differing economic conditions when they were active survey participants.

The first step of the functional principal component regression applies FPCA to the $X_{t+h|t-1}$, to obtain empirical functional principal components (EFPCs) and their corresponding scores. The second step replaces the term in $X_{t+h|t-1}$ in (4) and (5) by the scores, recasting the scalar-on-function regression in a multiple regression framework.

3.1. Functional principal component analysis

Given that FPCA may not be widely known, and to make our study self-contained, we provide a brief summary in the context of our study.

At time t, all the respondents' forecasts are collected, and the corresponding LQD of the cross-sectional distribution is obtained, denoted as $X_t(s), s \in (0, 1)$. $X_t(s)$ is assumed to be square integrable with the condition $\mathbb{E}\int X_t(s)ds < \infty$, and thus it can be regarded as a functional observation in the L^2 space. The sample $\{X_1(s), X_2(s), \ldots, X_T(s)\}$ is a sequence of functional observations, and can be viewed as the realizations of a random function $X(s)$. We can define the population mean and covariances of the random function $X(s)$ as:

$$\begin{aligned}
\mu(s) &= \mathbb{E}X(s), \\
c(s, s') &= \mathbb{E}\left[(X(s) - \mu(s))\left(X(s') - \mu(s')\right)\right].
\end{aligned}$$

Based on Mercer's lemma (Riesz et al., 1990), there exists an orthonormal sequence of continuous functions $v_k(s)$ in the sense that:

$$\int v_k(s)v_\ell(s)ds = \begin{cases} 0 & \text{if } k \neq \ell, \\ 1 & \text{if } k = \ell, \end{cases}$$

and a non-increasing sequence λ_k of positive numbers, such that:

$$c(s, s') = \sum_{k=1}^{\infty} \lambda_k v_k(s)v_k(s').$$

Typically, we obtain $v_k(s)$ as the eigenfunctions of the covariance function c, which are also called functional principal components (FPCs), and λ_k denotes the eigenvalues of c, in non-increasing order $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$. By the Karhunen–Loève expansion (Karhunen, 1947; Loève, 1960), one can decompose the random function $X(s)$ as:

$$X(s) = \mu(s) + \sum_{k=1}^{\infty} \xi_k v_k(s),$$

The random variables ξ_i are called the (principal component) scores, which are given by the projection of

$(X(s) - \mu(s))$ in the direction of the kth eigenfunction $v_k(s)$:

$$\xi_k = \int (X(s) - \mu(s))\, v_k(s) ds.$$

It can be shown that

$$\mathbb{E}\xi_k = 0, \qquad \mathbb{E}\xi^2 = \lambda_k, \qquad \mathrm{Cov}(\xi_k, \xi_\ell) = 0, \text{ if } k \neq \ell,$$

and

$$\mathbb{E} \int (X(s) - \mu(s))^2 \, ds = \sum_{k=1}^{\infty} \lambda_k.$$

Thus, it follows that λ_k is the variance of the random function $X(s)$ in the direction of the kth eigenfunction $v_k(s)$, and the sum of all λ_k is the total variance of $X(s)$.

In terms of estimation, the estimators of the population mean and covariance functions are the sample mean and covariance functions, defined by

$$\hat{\mu}(s) = \frac{1}{T} \sum_{t=1}^{T} X_t(s),$$

$$\hat{c}(s, s') = \frac{1}{T} \sum_{t=1}^{T} \left[\left(X_t(s) - \hat{\mu}(s) \right) \left(X_t(s') - \hat{\mu}(s') \right) \right].$$

The eigenfunctions of the sample covariance function $\hat{c}$ are the empirical functional principal components (EFPCs):

$$\hat{c}(s, s') = \sum_{k=1}^{\infty} \hat{\lambda}_k \hat{v}_k(s) \hat{v}_k(s'),$$

where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq 0$ are the sample eigenvalues of $\hat{c}(s, s')$. Then via the Karhunen–Loève expansion, the functional observations $\{X_1(s), \ldots, X_T(s)\}$ can be decomposed as:

$$\begin{aligned} X_t(s) &= \hat{\mu}(s) + \sum_{k=1}^{\infty} \hat{\xi}_{k,t} \hat{v}_k(s), \\ &= \hat{\mu}(s) + \sum_{k=1}^{K} \hat{\xi}_{k,t} \hat{v}_k(s) + e_t(s), \qquad t = 1, 2, \ldots, T \end{aligned} \tag{6}$$

where $\hat{\xi}_{k,t} = \int \left(X_t(s) - \hat{\mu}(s) \right) \hat{v}_k(s) ds$ is the estimated (principal component) score in the direction of the kth eigenfunction for the tth observation. When $K < T$ denotes the number of retained EFPCs, the $e_t(s)$ represents the truncation error function, and has zero mean and finite variance.

Eq. (6) illustrates the dimension reduction from infinity to a finite number K. This requires that the functional observation $X_t(s)$ can be well approximated from the first K EFPCs. That is, the information contained in functional observations $\{X_1(s), \ldots, X_T(s)\}$ can be effectively summarized by the finite-dimensional scores $\left\{ \hat{\boldsymbol{\xi}}_1, \ldots, \hat{\boldsymbol{\xi}}_T \right\}$, where $\hat{\boldsymbol{\xi}}_t = \left(\hat{\xi}_{1,t}, \ldots, \hat{\xi}_{K,t} \right)^\top$. These scores have zero mean and variance $\hat{\lambda}_k$, and they are uncorrelated with each other.

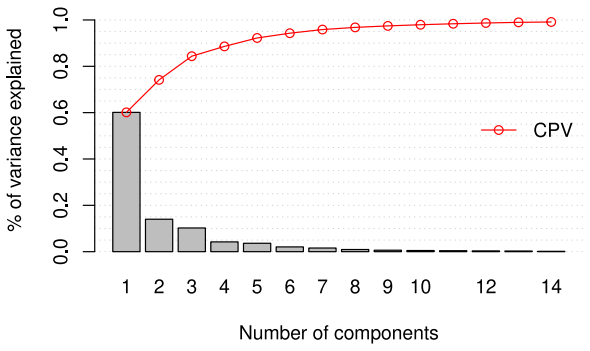


Fig. 2. CPV explained by the principal components for the pooled data.

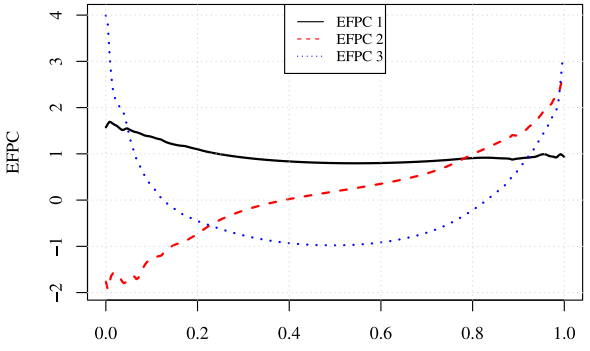


Fig. 3. EFPCs for the pooled cross-sectional distributions.

A practical question is how to choose K. Several criteria have been proposed: eigenvalue ratio tests (Ahn & Horenstein, 2013), cross-validation (Ramsay & Silverman, 2005), Akaike's information criterion (Akaike, 1974), the bootstrap method (Hall & Vial, 2006), and the cumulative percentage of variance (CPV) explained (Horváth & Kokoszka, 2012). CPV is commonly used (Bouri et al., 2021; Kearney & Shang, 2020; Shang & Kearney, 2022). The CPV explained by the first K EFPCs is computed by:

$$\mathrm{CPV}(K) = \frac{\sum_{k=1}^{K} \hat{\lambda}_k}{\sum_{k=1}^{T} \hat{\lambda}_k}.$$

We use the CPV method to select the number of K, and choose K such that at least 80% of the total variance is explained.

Empirical results of FPCA

Fig. 2 shows the variance explained by each of the principal components, and the cumulative variance, for the pooled (over h) data. Three EFPCs (i.e. $K = 3$) are sufficient to reach the threshold of explaining 80% of the total variance.

Fig. 3 shows the EFPCs for the pooled (over h) cross-sectional distributions, and Fig. 4 displays their corresponding scores for different horizons. The scores in Fig. 4 are calculated with respect to the common EFPCs obtained from the pooled data.

A challenge often encountered in FDA is how to interpret the FPCA, and in particular, the EFPCs. Similar issues

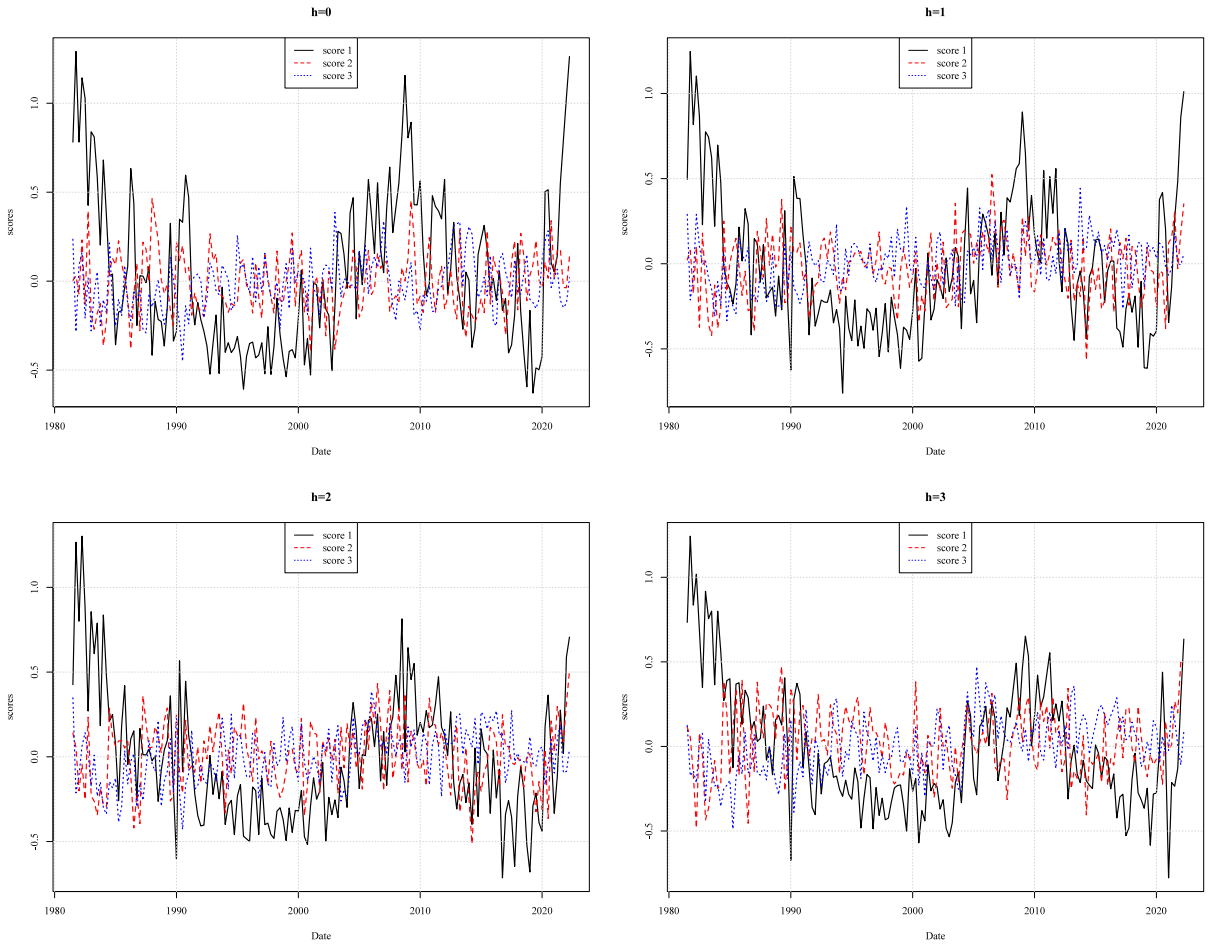


Fig. 4. Scores for the pooled cross-sectional distributions.

arise of course in standard principal component analysis. The EFPCs are orthonormal to one another, but may not be readily interpretable (Kokoszka & Reimherr, 2017). The conventional way of interpreting the scores is to consider their correlation (linear or rank-based) with other known factors or quantities. A relatively high correlation is generally taken to suppose a tentative interpretation of the score in terms of that factor or quantity.

Table 2 shows the estimates of Pearson's correlation and Spearman's rho between the first three estimated scores $\left(\hat{\xi}_{1,t}, \hat{\xi}_{2,t}, \hat{\xi}_{3,t}\right)$ and moments and descriptive statistics of the cross-sectional distributions,[14] viz. the mean, median, standard deviation (SD), interquartile range (IQR), skewness, and kurtosis. As can be observed, the first score $\hat{\xi}_{1,t}$ has a very high Pearson's correlation (0.94) with the SD measure of disagreement, and a slightly lower correlation of 0.83 with the IQR measure. The correlation between the second score $\hat{\xi}_{2,t}$ and the skewness is relatively high at 0.80. The third score $\hat{\xi}_{3,t}$ is more moderately correlated with the kurtosis (0.60). Thus, we conjecture that the three scores are related to the disagreement, skewness, and kurtosis.

In contrast to Pearson's correlation, which evaluates linear correlation, Spearman's rho evaluates monotonic relationships (whether linear or not). The Spearman's rho between $\hat{\xi}_{1,t}$ and SD is 0.96, between $\hat{\xi}_{2,t}$ and skewness is 0.86, and between $\hat{\xi}_{3,t}$ and kurtosis is 0.75. These are all higher than the corresponding linear correlations, and indicative of nonlinear relationships between the scores and the moments (SD, skewness, and kurtosis). Fig. 5 displays the scatter plots between the scores and the moments, and the nonlinearity can be clearly observed. The evidence supports the conjecture that the three scores measure disagreement, skewness, and kurtosis in a nonlinear manner.

It is possible that the cross-sectional distribution of inflation forecasts is correlated with other relevant variables, which are monitored by forecasters and influence expectations. One cannot rule out the possibility of omitted variables, but we consider two likely candidate variables in the case of inflation forecasting: crude oil prices

[14] We did not consider the shape factor used by Meeks and Monti (2023). This is because the calculation of the shape factor involves the pointwise time average distribution, which may not necessarily be in the density space, as described in Section 2.2.

Table 2
Correlation between EFPCs and moments.

	Pearson's correlation			Spearman's rho		
	Score 1	Score 2	Score 3	Score 1	Score 2	Score 3
Mean	0.32	0.10	−0.32	0.08	0.15	−0.33
Median	0.35	0.02	−0.32	0.09	0.06	−0.33
SD	**0.94**	0.00	−0.05	**0.96**	0.09	0.01
IQR	0.83	0.07	−0.48	0.78	0.12	−0.49
Skewness	−0.29	**0.80**	−0.06	−0.26	**0.86**	−0.05
Kurtosis	0.30	−0.06	**0.60**	0.38	−0.05	**0.75**
Oil price	0.28	0.07	0.30	0.35	0.06	0.30
Labor cost	−0.18	0.08	0.36	−0.10	0.05	0.37

Note: SD stands for standard deviation, and IQR stands for interquartile range.

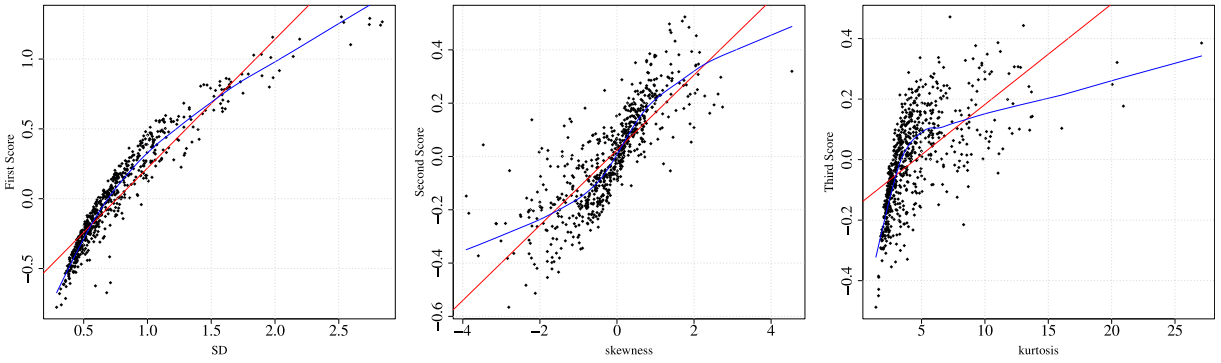


Fig. 5. Evidence of nonlinearity between the scores and the moments.

and labor costs.[15] The Pearson's correlations and Spearman's rho coefficients of these variables with the scores are shown in Table 2. We observe a moderate positive correlation between oil prices and both Score 1 and Score 3. This finding can be explained by elevated oil prices typically being associated with higher forecaster disagreement and potentially fatter tails (kurtosis) in the distribution of inflation forecasts. Additionally, labor costs have a modest correlation with Score 3, indicating that higher labor costs coincide with increased kurtosis in the distribution of inflation forecasts. These modest correlations suggest that much of the variation in the scores is not captured by these macro-variables.

3.2. Functional regression model

Having calculated the functional principal components, we next consider their employment in the Phillips curve regression models. Model (5) is a scalar-on-function functional regression: one of the regressors is a curve, and the dependent variable is scalar. Without imposing any regularity conditions, such a regression is infeasible due to the infinite dimension of the coefficient function $\gamma_j(\cdot)$. Reiss et al. (2017) provide a comprehensive review of the main methods for scalar-on-function regression. As they note, the main idea is to expand $\gamma_j(\cdot)$ in terms of a set of basis functions. The two types of basis functions are prior fixed bases, such as the B-spline basis and the Fourier basis; and data-driven bases, often derived by FPCA or functional partial least squares. If the basis is from FPCA, it is known as functional principal component regression (FPCR), which is widely used in the literature (e.g. Cao et al., 2020; Meeks & Monti, 2023). We choose to use FPCR because of its simplicity and interpretability (as discussed in Section 3.1).

Here, we briefly summarize FPCR. According to (6), we can have the approximation:

$$X_t(s) \approx \hat{\mu}(s) + \sum_{k=1}^{K} \hat{\xi}_{k,t}\hat{v}_k(s),$$
$$\text{where } \hat{\xi}_{k,t} = \int \left(X_t(s) - \hat{\mu}(s)\right)\hat{v}_k(s)ds.$$

Plugging this approximation into (5) gives:

$$E_{j,t}\pi_{t+h} = \zeta_j + \beta_j E_{j,t}\pi_{t+h+1} + \int \gamma_j(s)\left(\hat{\mu}(s) + \sum_{k=1}^{K}\hat{\xi}_{k,t}\hat{v}_k(s)\right)ds + \alpha_j E_{j,t}u_{t+h} + e_{j,t},$$

and we have

$$E_{j,t}\pi_{t+h} = \delta_j + \beta_j E_{j,t}\pi_{t+h+1} + \sum_{k=1}^{K}\hat{\xi}_{k,t}\gamma_{j,k} + \alpha_j E_{j,t}u_{t+h} + e_{j,t}, \quad (7)$$

[15] Specifically, we use the spot crude oil price of West Texas Intermediate (https://fred.stlouisfed.org/series/WTISPLC#) and unit labor costs for all workers in the nonfarm business sector (https://fred.stlouisfed.org/series/ULCNFB).

where

$$\delta_j = \zeta_j + \int \gamma_j(s)\hat{\mu}(s)ds, \qquad \text{and } \gamma_{j,k} = \int \gamma_j(s)\hat{v}_k(s)ds,$$

and α_j and β_j are regarded as unknown (scalar) parameters. Now, the functional regression model (5) is recast as a multiple regression with the scores replacing the curve. Then, we define the $T \times (K+3)$ matrix:

$$\boldsymbol{\Omega} = \begin{bmatrix} 1 & E_{j,1}\pi_{1+h} & \hat{\xi}_{1,1} & \hat{\xi}_{2,1} & \cdots & \hat{\xi}_{K,1} & E_{j,1}u_{1+h} \\ 1 & E_{j,2}\pi_{2+h} & \hat{\xi}_{1,2} & \hat{\xi}_{2,2} & \cdots & \hat{\xi}_{K,1} & E_{j,2}u_{2+h} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \\ 1 & E_{j,T}\pi_{T+h} & \hat{\xi}_{1,T} & \hat{\xi}_{2,T} & \cdots & \hat{\xi}_{K,T} & E_{j,T}u_{T+h} \end{bmatrix},$$

so that the parameter vector $\boldsymbol{\theta} = \left[\delta_j, \beta_j, \gamma_{j,1}, \ldots, \gamma_{j,K}, \alpha_j\right]^\top$ can be estimated by least squares multiple regression of $\boldsymbol{Y} = \boldsymbol{\Omega\theta} + \boldsymbol{e}$. If we denote the estimated parameters as $\hat{\delta}_j, \hat{\beta}_j, \hat{\gamma}_{j,1}, \ldots, \hat{\gamma}_{j,K}, \hat{\alpha}_j$, then the estimates of the parameters in the functional regression model (5) can be obtained as:

$$\hat{\gamma}_j(s) = \sum_{k=1}^{K} \gamma_{j,k}\hat{v}_k(s), \quad \text{and} \quad \hat{\zeta}_j = \hat{\delta}_j - \sum_{k=1}^{K} \hat{\gamma}_{j,k} \int \hat{v}_k(s)\hat{\mu}(s)ds. \tag{8}$$

Empirical results of FPCR

There are in total 256 unique individual respondents in the U.S. SPF dataset for the period we consider.[16] In order to ensure sufficient forecasts to obtain reasonably precise estimates at an individual level, we select only the respondents who made 120 or more forecasts (for all $h = 0, 1, 2, 3$) which is at least 30 forecasts per h on average. This gave 64 respondents. Then we undertook FPCR for each respondent j, pooling over h for that j.

To summarize the results, Fig. 6 provides a boxplot of the parameter estimates for Model (7) for each respondent, and the top panel of Table 3 presents the summary statistics of parameter estimates across respondents. We report the cross-sectional mean, standard deviation, lower quartile, median, and upper quartile for each of the estimated parameters. We also report the proportion of respondents for whom we reject the null that the coefficient is zero, at the 1%, 5%, and 10% levels. Additionally, we report summary statistics for the adjusted R^2 and number of observations.[17] The bottom panel of Table 3 reports the parameter estimates when the scores are excluded from the model.

Consider the top panel first. Nearly half of all respondents (45%) appear to use the information in Score 1 when they produce their forecasts. Since Score 1 is a (nonlinear) measure of disagreement, it appears that disagreement at $t-1$ is an important determinant of around half of the respondents' forecasts made at time t (of $t+h$). Forecaster disagreement has often been regarded as a proxy for perceived (or actual) uncertainty about the future outlook, and may be playing a similar role in our analysis (see e.g. Zarnowitz and Lambros (1987) and Haddow et al. (2013)). For the majority of agents, $\hat{\gamma}_{j,1} > 0$, suggesting that higher inflation disagreement at time $t-1$ results in an increase in j's forecast made at time t. Empirically this seems plausible because inflation rates and inflation variability have often been found to be positively correlated. The interpretation would be that if inflation disagreement last period $(t-1)$ was high, the majority of respondents would increase their time t forecasts of inflation in $t+h$ by more than would be suggested by their own forecasts of inflation in $t+h+1$ and the unemployment rate in $t+h$.

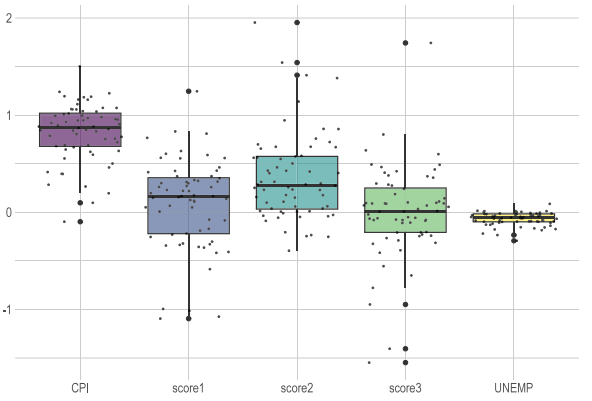


Fig. 6. Boxplot of the parameter estimates.

Secondly, a modest percentage of all respondents (19% at a 5% significance test level) employ Score 2 in the cross-sectional distribution of all forecasts at $t-1$. Recall that Score 2 nonlinearly measures the skewness in the distribution. Thus, we find evidence that some respondents take skewness into consideration when they generate their forecasts. However, the effect of a change in skewness is less amenable to a behavioral interpretation. The parameter estimates of Score 2 are positive for most agents, albeit not statistically significantly different from zero. In Section 4 we attempt to shed further light on the effects of the skew, by allowing the effects of positive and negative skew to differ.

Thirdly, a small portion of respondents (8% at the 5% level) incorporate the information in Score 3 for their forecasts. As discussed, Score 3 is nonlinearly related to the kurtosis of the distribution, which can possibly capture a fat-tailed forecast distribution.

As described above, FDA provides a number of insights into how respondents' forecasts are influenced by those of others. An overall test of the significance of the FPCA scores in the FPCRs is one way of assessing the importance of applying FDA to the analysis of macro-survey expectations. We calculated an F-test of $\gamma_{j,1} = \gamma_{j,2} = \gamma_{j,3} = 0$ for each respondent. We rejected the null for nearly half of the respondents at the 5% level. Hence, we find that nearly half of the respondents use information contained in the distribution of other individuals' forecasts.

[16] The respondents are anonymous in the U.S. SPF, but each has a unique identifier, allowing for an analysis of all the forecasts associated with a given identifier.

[17] Since we pool all h, the number of observations for one respondent is required to be at least 120.

Table 3
Summary of parameter estimates of Model (7) across respondents.

With Scores 1–3								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.758	0.749	0.333	0.816	1.073	46.9%	60.9%	75.0%
CPI	0.825	0.300	0.679	0.875	1.021	95.3%	96.9%	96.9%
Score 1	0.092	0.462	−0.220	0.167	0.356	28.1%	46.9%	56.3%
Score 2	0.364	0.454	0.033	0.280	0.576	6.3%	18.8%	28.1%
Score 3	0.000	0.475	−0.206	0.012	0.250	4.7%	7.8%	10.9%
UNEMP	−0.068	0.069	−0.098	−0.057	−0.016	32.8%	39.1%	45.3%
F-test on Scores 1–3						25.0%	45.3%	53.1%
Adj. R^2	46.8%	25.0%	28.5%	46.9%	63.7%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			
Without Scores 1–3								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.652	0.783	0.059	0.634	1.063	45.3%	62.5%	68.8%
CPI	0.848	0.302	0.704	0.895	1.056	95.3%	96.9%	98.4%
UNEMP	−0.062	0.062	−0.096	−0.057	−0.011	28.1%	43.8%	50.0%
Adj. R^2	45.4%	25.6%	26.6%	44.4%	63.9%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

If the scores are excluded, that is, if the influence of others is ruled out, the cross-sectional distribution of unemployment rate coefficients is little changed, as shown in the bottom panel of Table 3. The views of others significantly influence nearly half of the respondents, but their omission does not markedly affect the estimates of the unemployment rate coefficient (the average value changes from −0.068 to −0.062).

Characteristics of respondents

Of interest is whether we are able to identify the characteristics of respondents which are correlated with the tendency to draw on information contained in (the distribution of) the forecasts of others. To investigate this question, we firstly select five characteristics which might potentially be relevant. They include measures of various aspects of the strength of the Phillips curve 'imprint' on the forecasts,[18] that is, the extent to which an individual's inflation and unemployment rate forecasts reveal a belief that higher 'slack' in the economy puts downward pressure on inflation. The characteristics are:

- $\hat{\alpha}_j$: Respondent j's estimated parameter of the unemployment rate in Model (5),
- sign($\hat{\alpha}_j$): the sign of $\hat{\alpha}_j$,
- $\hat{\beta}_j$: Respondent j's estimated parameter of the inflation rate in Model (5),
- Period$_j$: the proportion of Respondent j's forecasts made in the earlier two decades (1981:Q3–1999:Q4) of the sample period,
- $\tilde{\sigma}_j$: an uncertainty measure of Respondent j's forecasts across $h = 0, 1, 2, 3$.

Hence, the first three characteristics are based on the individual's estimates of Model (5), and capture how respondent j believes future inflation is related to the unemployment rate. Following Clements (2024), we include an indicator (Period$_j$) of whether the respondent was primarily active earlier or later in the period. Higher values of Period$_j$ indicate that the respondent made a larger proportion of forecasts in the earlier period (1981:Q3–1999:Q4). A value of zero indicates that an individual was only active in the 21st century (2000:Q1 to 2022:Q3). We wish to allow for the possibility that the inflation process might have changed over the nearly 40-year period we consider, with a concomitant change in forecaster behavior. Finally, the fifth characteristic is a measure of the uncertainty of Respondent j's forecasts, calculated by $\tilde{\sigma}_j = \frac{1}{4}\sum_{h=0}^{3}\sqrt{\text{Var}(E_{j,t}\pi_{t+h})}$. This measures the volatility of Respondent j's forecasts.[19]

Table 4 shows summary statistics (the mean, standard deviation, and the number of respondents) of the five respondent-specific characteristics separately for the

[18] Recall that the respondents are anonymous, precluding analyzing the effects of personal characteristics.

[19] An alternative would be to use the respondents' histogram forecasts to generate measures of perceived uncertainty, as described by Clements et al. (2022). Complications would arise in doing so, in particular because the histograms are fixed-event and target the annual year-on-year growth rates, in contrast to the fixed-horizon quarterly point forecasts, but see Ganics et al. (2023) on deriving series of fixed-horizon forecasts from fixed-event forecasts. This is potentially interesting because one might conjecture that an individual's perceptions of uncertainty will vary over time and that more weight would be accorded to the forecasts of others at times of high perceived uncertainty. We do not pursue this here, but we consider whether reliance on others varies across the business cycle (with expansions/contractions being correlated with low/high uncertainty), and in Appendix D, we check whether our findings are robust to the inclusion of VIX as a general measure of uncertainty.

Table 4
Difference between respondent characteristics.

			$\hat{\alpha}_j$	sign($\hat{\alpha}_j$)	$\hat{\beta}_j$	$Period_j$	$\tilde{\sigma}_j$
Score 1	Significant group	mean	−0.08	−0.80	0.86	0.21	1.03
		sd	*(0.07)*	*(0.61)*	*(0.32)*	*(0.28)*	*(0.30)*
		no. of respondents	30	30	30	30	30
	Insignificant group	mean	−0.06	−0.71	0.80	0.33	1.13
		sd	*(0.06)*	*(0.72)*	*(0.28)*	*(0.39)*	*(0.32)*
		no. of respondents	34	34	34	34	34
	Two-sample t-test	p-value	31.8%	57.3%	42.8%	13.9%	21.1%
Score 2	Significant group	mean	−0.05	−1.00	0.64	0.23	1.16
		sd	*(0.05)*	*(0.00)*	*(0.25)*	*(0.27)*	*(0.29)*
		no. of respondents	12	12	12	12	12
	Insignificant group	mean	−0.07	−0.69	0.87	0.28	1.07
		sd	*(0.07)*	*(0.73)*	*(0.30)*	*(0.36)*	*(0.32)*
		no. of respondents	52	52	52	52	52
	Two-sample t-test	p-value	31.8%	0.4%	1.4%	61.2%	33.2%
Score 3	Significant group	mean	−0.09	−1.00	0.80	0.37	1.19
		sd	*(0.10)*	*(0.00)*	*(0.29)*	*(0.39)*	*(0.43)*
		no. of respondents	5	5	5	5	5
	Insignificant group	mean	−0.07	−0.73	0.83	0.27	1.07
		sd	*(0.07)*	*(0.69)*	*(0.30)*	*(0.34)*	*(0.30)*
		no. of respondents	59	59	59	59	59
	Two-sample t-test	p-value	70.9%	0.4%	82.3%	60.4%	57.0%
F-test on Scores 1–3	Significant group	mean	−0.08	−0.93	0.79	0.25	1.08
		sd	*(0.06)*	*(0.37)*	*(0.28)*	*(0.29)*	*(0.29)*
		no. of respondents	29	29	29	29	29
	Insignificant group	mean	−0.06	−0.60	0.85	0.29	1.09
		sd	*(0.08)*	*(0.81)*	*(0.32)*	*(0.39)*	*(0.34)*
		no. of respondents	35	35	35	35	35
	Two-sample t-test	p-value	33.2%	3.6%	44.9%	65.7%	89.6%

Note: This table shows the mean, the standard deviation, and the number of respondents of the five respondent characteristics in the significant group and the insignificant group, according to Score 1, Score 2, and Score 3, individually and jointly. The last row in each panel of this table shows the p-value of the Welch two-sample t-test.

'significant' and 'insignificant' groups, according to each of Score 1, Score 2, and Score 3 (and jointly). That is, for Score 1 we separate the respondents into the group for whom we reject the null that Score 1 is statistically insignificant at the 5% level, and a group comprising those for whom we do not reject it. We then consider whether each of the five characteristics differs between the two groups. And then we repeat for Score 2, etc. The last row in each panel of Table 4 presents the p-value of the Welch two-sample t-test. Based on this test, there is no significant difference in the mean of $\hat{\alpha}_j$ and $\tilde{\sigma}_j$. However, we find strong evidence that the mean of sign($\hat{\alpha}_j$) is higher in the 'significant group' for Score 2, Score 3, and for the group for which we reject for all the scores together. This indicates a positive association between believing that higher economic slack reduces inflation pressure and paying heed to the forecasts of others (at least as measured by Scores 2 and 3). Put differently, a stronger belief in the Phillips curve (measured by a negative unemployment rate coefficient in the Phillips curve model) is correlated with being influenced by the forecasts of others. It is not obvious why 'theory-consistency' and 'paying attention to others' should go together, and this merits further investigation.

We find little evidence that $\hat{\beta}_j$ varies across the significant and insignificant groups for the Scores, except for Score 2 (with a 1.4% p-value for the two-sample t-test).

4. Robustness checks

We made a number of modeling choices in deriving the results in Section 3. In this section we consider the robustness of those findings to various alternative choices and extensions, with a key emphasis on whether the application of FDA provides additional insight into forecaster behavior relative to the use of non-FDA methods. To this end, the first set of checks includes the consensus forecast, i.e. the mean of the cross-sectional distribution forecasts, in the individual Phillips curve models (Section 4.1), and we also include the higher cross-sectional moments (Section 4.2).

Next, we allow for the possibility that a simple linear relationship may not hold over the whole sample period, as suggested by Fendel et al. (2011) and others (albeit (Fendel et al., 2011) consider consensus forecasts as opposed to forecasts at the level of the individual). We do this in two ways. Firstly, in Section 4.3 we allow for the possibility that the relationship differs between normal and abnormal times, where these descriptors are made operational as NBER-dated expansions and contractions. Secondly, in Section 4.4 we allow for the possibility that forecaster behavior might be different in high inflation and low inflation times as well as whether inflation is close to, or far from, the Federal Reserve's target rate. Finally, some authors suggest modeling inflation in gaps

(see e.g. Faust and Wright (2013)), and we check the robustness of our findings to adopting such a formulation for respondents' inflation forecasts in Section 4.5.

In the appendix, we provide robustness checks in addition to those included here: namely, we replace the expectations-augmented Phillips curve with a hybrid Phillips curve model; decompose Score 2; consider the effects of allowing horizon fixed effects; of including VIX in the Phillips curve models; of specifying the Phillips curve models with the unemployment rate gap; and we consider the treatment of outliers.

4.1. Including the cross-sectional mean of the inflation forecasts

A potential issue with our analysis is that we did not include the consensus forecast in the models for each individual's inflation forecasts. The consensus is often taken to be the (cross-sectional) mean. Our FPCA scores are not highly correlated with the cross-sectional mean of the inflation forecasts (see Table 2), and hence would not appear to be simply capturing the effect of the mean.[20] Nevertheless, to determine whether its exclusion accounts for the significance of the scores (e.g. if the disagreement and the level of inflation were highly correlated), we run:

$$E_{j,t}\pi_{t+h} = \delta_j + \beta_j E_{j,t}\pi_{t+h+1} + \theta_j M_{t+h|t-1} + \sum_{k=1}^{3} \hat{\xi}_{k,t}\gamma_{j,k} + \alpha_j E_{j,t}u_{t+h} + e_{j,t}, \quad (9)$$

where

$$M_{t+h|t-1} = \frac{1}{N_{t-1}} \sum_{j=1}^{N_{t-1}} E_{j,t-1}\pi_{t+h},$$

is the cross-sectional mean, and N_{t-1} is the number of forecasts made at $t-1$ for π_{t+h}.

Table 5 presents the summary statistics (across j) for Model (9) estimated for each respondent. The percentage of F-test rejections is a little lower, at 36% compared to 45%, but clearly the exclusion of the mean does not account for the relevance of the scores for most respondents.

4.2. Including higher cross-sectional moments of the inflation forecasts

Our second robustness check acknowledges that the FPCA scores are highly correlated (whether linearly or not) with the higher moments (SD, skewness, and kurtosis) of the cross-sectional distribution of the inflation forecasts in Section 3.1. Hence, we examine the extent to which the scores contain additional information, after controlling for the higher moments. To this end, we run the regression:

$$E_{j,t}\pi_{t+h} = \delta_j + \beta_j E_{j,t}\pi_{t+h+1} + \psi_j mom_{t+h|t-1} + \sum_{k=1}^{3} \hat{\xi}_{k,t}\gamma_{j,k} + \alpha_j E_{j,t}u_{t+h} + e_{j,t}, \quad (10)$$

where $mom_{t+h|t-1}$ corresponds to one of the higher cross-sectional moments (SD, skewness, and kurtosis) of the forecasts made at $t-1$ for π_{t+h}.

Table 6 provides the summary statistics for Model (10) for each respondent. Firstly, we observe that SD is statistically significant (at the 5% level) for 41% of respondents, which is perhaps as expected given that SD is a standard linear measure of forecaster disagreement. Despite Score 1 being highly correlated with SD, it is still statistically significant (at the 5% level) for 25% of the respondents, indicating it contains useful additional information even after controlling for SD. Secondly, the percentage for whom Score 2 is statistically significant is a little reduced (at 13% from 19%), after controlling for the cross-sectional skewness. Thirdly, the cross-sectional kurtosis is generally not significant (the percentages of rejections are less than significance levels). Lastly, the F-test rejection rate (of the three scores taken together) is reduced to 25% after controlling for SD, marginally reduced to 37.5% after controlling for skewness, and remains the same at 45.3% after controlling for kurtosis.

Table 7 provides a different comparison. It estimates the basic model but with the three moments replacing the three scores in Table 3. The rejection rates across the two tables for the moments and scores respectively (i.e. of the variables capturing the influence of others' forecasts) are very similar, as are the estimates of the cross-sectional distribution of unemployment rate coefficients. We postpone formal encompassing tests of the two models to Section 4.3.

4.3. Normal versus abnormal times

We define abnormal times as the quarter before the onset of a recession, the recessionary period itself, and the quarter following the end of a recession, where recessions are dated from NBER US Business Cycle Expansions and Contractions.[21] Other periods are designated as normal times. Abnormal times are typically periods of high uncertainty.[22] We run our baseline regression of Model (7) separately for the two sub-periods, and the results are presented in Table 8. Our findings suggest that the responsiveness of inflation to unemployment is greater in high-uncertainty (recessionary) periods: the mean coefficient on the unemployment rate is −0.188, compared to −0.043 in low-uncertainty (expansionary) periods. However, the FDA scores have a greater impact in low-uncertainty normal times. The rejection of the F-test (for Scores 1–3) is 50.0% for the low economic uncertainty sub-period, and 12.5% for the sub-period of high economic uncertainty. In normal times, forecasters are more attentive to others and put less weight on their own

[20] There are a number of reasons why one might in principle expect the mean to be a relevant variable: see for example Clements (2018).

[21] https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions.

[22] In Appendix D, we also check whether our findings are robust to an uncertainty measure relating to broad economic conditions.

Table 5
Summary of parameter estimates of Model (9).

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	−0.200	1.037	−0.746	−0.032	0.358	18.8%	32.8%	40.6%
CPI	0.706	0.288	0.541	0.731	0.921	92.2%	93.8%	95.3%
Mean	0.499	0.433	0.183	0.400	0.756	54.7%	73.4%	79.7%
Score 1	0.099	0.454	−0.158	0.178	0.327	26.6%	34.4%	48.4%
Score 2	0.049	0.458	−0.185	0.039	0.243	4.7%	9.4%	15.6%
Score 3	−0.004	0.444	−0.164	0.016	0.223	3.1%	7.8%	9.4%
UNEMP	−0.055	0.082	−0.099	−0.053	−0.004	23.4%	42.2%	51.6%
F-test on Scores 1–3						18.8%	35.9%	40.6%
Adj. R^2	49.3%	24.3%	29.1%	50.8%	66.6%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Table 6
Summary of parameter estimates of Model (10).

Controlling SD								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.977	1.233	0.270	0.813	1.657	37.5%	48.4%	53.1%
CPI	0.809	0.289	0.677	0.849	1.013	95.3%	96.9%	96.9%
SD	−0.279	1.435	−1.210	−0.131	0.611	18.8%	40.6%	46.9%
Score 1	0.325	0.881	−0.239	0.268	0.971	9.4%	25.0%	35.9%
Score 2	0.386	0.473	0.033	0.297	0.654	10.9%	26.6%	32.8%
Score 3	0.092	0.556	−0.148	0.003	0.405	4.7%	15.6%	21.9%
UNEMP	−0.066	0.068	−0.099	−0.051	−0.011	32.8%	39.1%	45.3%
F-test on Scores 1–3						14.1%	25.0%	42.2%
Adj. R^2	47.6%	24.7%	29.8%	47.6%	64.1%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			
Controlling skewness								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.757	0.759	0.302	0.820	1.079	45.3%	62.5%	75.0%
CPI	0.824	0.303	0.680	0.868	1.030	95.3%	96.9%	98.4%
Skewness	−0.036	0.148	−0.108	−0.036	0.002	1.6%	7.8%	12.5%
Score 1	0.062	0.459	−0.187	0.131	0.388	18.8%	39.1%	45.3%
Score 2	0.516	0.767	0.158	0.385	0.986	0.0%	12.5%	23.4%
Score 3	−0.025	0.467	−0.220	0.018	0.236	4.7%	9.4%	10.9%
UNEMP	−0.068	0.070	−0.100	−0.055	−0.015	32.8%	39.1%	45.3%
F-test on Scores 1–3						18.8%	37.5%	48.4%
Adj. R^2	46.8%	24.8%	28.3%	46.7%	64.0%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			
Controlling kurtosis								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.740	0.742	0.291	0.772	1.052	40.6%	57.8%	65.6%
CPI	0.824	0.300	0.679	0.878	1.025	95.3%	96.9%	96.9%
Kurtosis	0.008	0.026	−0.003	0.006	0.017	1.6%	3.1%	9.4%
Score 1	0.078	0.455	−0.215	0.170	0.347	21.9%	37.5%	53.1%
Score 2	0.365	0.452	0.027	0.278	0.592	7.8%	20.3%	29.7%
Score 3	−0.075	0.649	−0.383	−0.074	0.188	3.1%	12.5%	17.2%
UNEMP	−0.071	0.070	−0.107	−0.056	−0.018	32.8%	42.2%	48.4%
F-test on Scores 1–3						28.1%	45.3%	51.6%
Adj. R^2	46.7%	25.0%	28.3%	46.6%	64.1%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Table 7
Summary of parameter estimates of the model with cross-sectional moments only.

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.666	0.803	0.071	0.590	1.165	39.1%	48.4%	57.8%
CPI	0.818	0.295	0.672	0.880	1.011	95.3%	96.9%	98.4%
SD	0.122	0.706	−0.368	0.232	0.505	37.5%	59.4%	65.6%
Skew	0.047	0.088	−0.005	0.035	0.075	4.7%	14.1%	17.2%
Kurtosis	0.005	0.024	−0.010	0.002	0.019	3.1%	4.7%	15.6%
UNEMP	−0.066	0.069	−0.108	−0.051	−0.015	28.1%	42.2%	50.0%
F-test on SD, skew, kurtosis						26.6%	45.3%	56.3%
Adj. R^2	47.0%	24.9%	27.9%	47.0%	64.7%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Table 8
Summary of parameter estimates of Model (7) across respondents during high/low economic uncertainty.

High economic uncertainty								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	1.607	3.346	0.000	0.924	2.170	17.2%	29.7%	35.9%
CPI	0.814	0.598	0.531	0.892	1.158	65.6%	73.4%	81.3%
Score 1	−0.222	1.311	−0.587	−0.037	0.511	3.1%	20.3%	32.8%
Score 2	−0.021	1.109	−0.592	−0.010	0.684	0.0%	6.3%	10.9%
Score 3	0.838	1.595	−0.047	0.909	1.862	1.6%	7.8%	14.1%
UNEMP	−0.188	0.441	−0.242	−0.093	−0.013	21.9%	34.4%	45.3%
F-test on Scores 1–3						3.1%	12.5%	14.1%
Adj. R^2	47.7%	26.6%	28.0%	46.2%	69.2%			
No. of obs.	46.0	18.5	35.0	48.0	60.0			
Low economic uncertainty								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.811	0.841	0.270	0.739	1.179	50.0%	54.7%	62.5%
CPI	0.741	0.360	0.602	0.805	0.992	85.9%	87.5%	89.1%
Score 1	0.146	0.408	−0.076	0.172	0.423	28.1%	39.1%	46.9%
Score 2	0.438	0.478	0.088	0.305	0.655	15.6%	31.3%	35.9%
Score 3	−0.015	0.516	−0.282	0.007	0.240	3.1%	14.1%	14.1%
UNEMP	−0.043	0.060	−0.090	−0.040	−0.005	18.8%	34.4%	42.2%
F-test on Scores 1–3						32.8%	50.0%	59.4%
Adj. R^2	46.0%	27.5%	24.8%	47.7%	65.9%			
No. of obs.	197.5	74.2	132.0	180.0	253.0			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$) during sub-periods of high/low economic uncertainty, for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

unemployment rate forecasts. Given that normal-time observations outnumber abnormal times approximately 4:1, the former periods hold greater sway and drive the baseline findings for the whole period. The lower number of recessionary observations on average is of course at least a partial explanation of the differences in statistical significance (e.g. of the *F*-test of Scores 1–3).

Given the differences between the results for the two sets of periods, we repeat the exercise with the three moments replacing the functional factors, and report the results in Table 9. In abnormal periods, the rejection percentages of the moments-based model have a similar pattern to the model with scores (Table 8). However, in normal periods there are some noticeable differences: Score 1 (39.1%) is rejected less often than SD (46.9%), Score 2 (31.3%) is rejected over twice as often as skewness (14.1%), and Score 3 (14.1%) is rejected twice as often as kurtosis (6.3%). This suggests there is not a one-to-one correspondence between the moment and functional factor formulations once we allow for differences across business-cycle phases.

Following Meeks and Monti (2023), we report the Davidson and MacKinnon (1981) *J*-test of FPCR (Model (7)) and the moment-based model, both for the whole period, and for normal and abnormal times separately. In Section 4.3 we found that the two models appeared to perform similarly over the whole period, in terms of the distribution of unemployment rate coefficient estimates and the influence of others' forecasts (compare Tables 3 and 7). Table 10 formally tests the two models. It

Table 9
Summary of parameter estimates of the moments-based model across respondents during high/low economic uncertainty.

High economic uncertainty								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	1.710	3.289	−0.028	1.010	2.154	15.6%	31.3%	39.1%
CPI	0.822	0.624	0.528	0.930	1.141	62.5%	70.3%	78.1%
SD	−0.302	1.818	−0.997	−0.152	0.616	6.3%	25.0%	40.6%
Skew	0.060	0.292	−0.064	0.084	0.199	0.0%	6.3%	10.9%
Kurtosis	0.007	0.168	−0.014	0.040	0.090	6.3%	12.5%	15.6%
UNEMP	−0.184	0.410	−0.210	−0.107	−0.021	20.3%	34.4%	37.5%
F-test on Scores 1–3						3.1%	14.1%	26.6%
Adj. R^2	49.1%	25.2%	30.0%	46.1%	70.6%			
No. of obs.	46.0	18.5	35.0	48.0	60.0			
Low economic uncertainty								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.658	0.923	−0.056	0.465	1.282	35.9%	45.3%	59.4%
CPI	0.735	0.353	0.605	0.812	0.986	85.9%	89.1%	89.1%
SD	0.299	0.628	−0.102	0.270	0.753	31.3%	46.9%	50.0%
Skew	0.049	0.078	0.008	0.030	0.093	6.3%	14.1%	21.9%
Kurtosis	−0.005	0.030	−0.017	−0.005	0.010	1.6%	6.3%	14.1%
UNEMP	−0.046	0.059	−0.087	−0.050	−0.012	20.3%	34.4%	40.6%
F-test on Scores 1–3						28.1%	46.9%	50.0%
Adj. R^2	45.9%	27.7%	24.2%	48.2%	67.1%			
No. of obs.	197.5	74.2	132.0	180.0	253.0			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$) during sub-periods of high/low economic uncertainty, for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Table 10
Summary of the Davidson and MacKinnon (1981) J-test across respondents.

	rej. 1%	rej. 5%	rej. 10%
Full-sample			
FPCR encompasses the moments-based model	6.3%	25.0%	32.8%
The moments-based model encompasses FPCR	26.6%	42.2%	42.2%
Both models encompass each other	1.6%	1.6%	6.3%
Neither model encompasses the other	65.6%	31.3%	18.8%
High economic uncertainty			
FPCR encompasses the moments-based model	0.00%	3.13%	14.06%
The moments-based model encompasses FPCR	4.69%	26.56%	31.25%
Both models encompass each other	0.00%	1.56%	3.13%
Neither model encompasses the other	95.31%	68.75%	51.56%
Low economic uncertainty			
FPCR encompasses the moments-based model	12.5%	28.1%	32.8%
The moments-based model encompasses FPCR	12.5%	21.9%	29.7%
Both models encompass each other	0.0%	1.6%	6.3%
Neither model encompasses the other	75.0%	48.4%	31.3%

shows the percentage of respondents for whom the J test indicates that the FPCR encompasses the moment-based model, the percentage for whom the moment-based model encompasses the FPCR, the percentage for whom both models encompass the other, and the percentage of respondents for whom neither model encompasses the other.

Across respondents, it appears that the functional factors and cross-sectional moments carry different information—in the sense that only for a few respondents do we find that both models encompass the other. If each model encompasses the other then we might suppose that the functional factors and cross-sectional moments are effectively equivalent ways of capturing the impact of the forecasts of others (for those respondents). If we consider the full sample, we find that FPCR encompasses the moments-based model for 25% of all respondents, whilst in the opposite direction, the moments-based model encompasses FPCR for 42.2% of respondents (at the 5% significance level). In abnormal times, the moments-based model encompasses FPCR more often than the other way round, but this is reversed in normal times. In the two sub-periods, the most common outcome is that neither model encompasses the other, suggesting heterogeneity in the way in which forecasters respond to others which is not adequately captured by either the functional factors or moments formulations.

Table 11
Summary of parameter estimates of Model (7) across respondents during high/low inflation times.

High inflation times								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	1.150	0.910	0.520	0.985	1.791	56.3%	64.1%	73.4%
CPI	0.751	0.310	0.584	0.825	0.943	85.9%	92.2%	92.2%
Score 1	0.344	0.391	0.114	0.361	0.596	32.8%	51.6%	59.4%
Score 2	0.259	0.553	−0.025	0.148	0.460	3.1%	9.4%	15.6%
Score 3	−0.295	0.632	−0.537	−0.229	−0.042	9.4%	17.2%	20.3%
UNEMP	−0.084	0.070	−0.132	−0.075	−0.031	28.1%	42.2%	48.4%
F-test on Scores 1–3						32.8%	46.9%	56.3%
Adj. R^2	49.2%	26.4%	26.5%	50.6%	71.9%			
No. of obs.	167.6	68.9	108.0	156.0	219.0			
Low inflation times								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.267	1.542	−0.383	0.031	1.200	20.3%	26.6%	39.1%
CPI	0.775	0.584	0.461	0.794	1.109	67.2%	73.4%	81.3%
Score 1	−0.538	1.118	−1.148	−0.539	−0.026	29.7%	35.9%	51.6%
Score 2	0.529	1.114	−0.077	0.328	0.815	9.4%	12.5%	18.8%
Score 3	0.884	2.380	0.080	0.469	0.905	3.1%	9.4%	10.9%
UNEMP	−0.011	0.163	−0.085	−0.013	0.073	15.6%	28.1%	31.3%
F-test on Scores 1–3						23.4%	35.9%	42.2%
Adj. R^2	41.2%	23.5%	25.9%	38.1%	55.3%			
No. of obs.	75.8	35.0	52.0	78.0	98.5			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

4.4. High versus low inflation times

As well as supposing that forecaster behavior may differ between recessionary and expansionary periods, the extent to which forecasters look at others' forecasts may depend on whether inflation is high. Higher inflation suggests increased attentiveness to news, and this may result in greater attention to others' forecasts as a source of information. We address this question by investigating the difference in the parameter estimates of Model (7) between high and low inflation times. We define high inflation times as when the CPI inflation rate headline is more than 2% (annualized, quarter-over-quarter percentage change), based on the CPI data from FRED.[23]

Table 11 reports the estimates of running the baseline regression separately for the two sub-periods. Relative to the whole-sample baseline results in Table 3, one notable difference is that more respondents (17.2%) have a significant coefficient on Score 3 during high inflation times, where Score 3 is a nonlinear measure of kurtosis. In addition, the mean coefficient on Score 3 during high inflation times is −0.295, implying that respondents make more conservative inflation forecasts during high inflation periods when the distribution of others' forecasts has fatter tails. Another notable finding is that the mean coefficient on Score 1 is 0.344 during high inflation times, but −0.538 during low inflation times (compared to a full sample estimate of 0.092). This suggests that forecast disagreement has a greater effect during high inflation times—compared to the full sample estimate—and that greater forecast disagreement reduces respondents' forecasts made during low inflation times. Lastly, based on the F-test, more respondents (46.9%) use information contained in the distribution of others' forecasts during higher inflation times, compared to 35.9% of respondents during low inflation times.

We also investigate whether the response to others' forecasts depends on whether inflation is close to, or far from, the target rate. The Federal Reserve has maintained a long-run inflation target of 2% at an annualized rate. Thus, we define the sub-period of inflation being close to the target as the periods when inflation is between 1% and 3%, and far when inflation is outside this interval. The results are shown in Table 12. Compared to the whole-sample baseline results in Table 3, there is a substantial reduction in the number of respondents who have significant F-test statistics during the sub-period when inflation is close to the target, to around 10% of the number of respondents, but only a modest reduction during the sub-period when inflation is far from the target. Our results suggest that forecasters pay little attention to the forecasts of others when inflation is close to the target.

4.5. Regression based on the inflation gap

Our inflation-gap formulation of the Phillips curve model is:

$$E_{j,t}\pi_{t+h} - E_{j,t}\tau_t = \delta_j + \beta_j(E_{j,t}\pi_{t+h+1} - E_{j,t}\tau_t) + \sum_{k=1}^{3}\hat{\xi}_{k,t}\gamma_{j,k} + \alpha_j E_{j,t}u_{t+h} + e_{j,t}, \quad (11)$$

[23] https://fred.stlouisfed.org/series/CPIAUCSL#.

Table 12
Summary of parameter estimates of Model (7) across respondents during inflation times close to, or far from, the target rate.

Inflation close to the target rate								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.964	0.807	0.389	0.794	1.389	40.6%	62.5%	64.1%
CPI	0.633	0.308	0.455	0.725	0.865	81.3%	84.4%	87.5%
Score 1	−0.054	0.361	−0.194	−0.003	0.181	4.7%	15.6%	21.9%
Score 2	0.211	0.588	−0.059	0.151	0.350	3.1%	7.8%	10.9%
Score 3	0.170	0.689	−0.194	0.146	0.458	0.0%	1.6%	7.8%
UNEMP	−0.032	0.111	−0.064	−0.039	−0.004	6.3%	18.8%	32.8%
F-test on Scores 1–3						1.6%	9.4%	12.5%
Adj. R^2	43.0%	24.9%	27.2%	42.8%	58.7%			
No. of obs.	111.8	50.5	78.8	100.0	141.0			
Inflation far from the target rate								
	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.691	0.911	0.140	0.702	1.172	23.4%	40.6%	45.3%
CPI	0.895	0.333	0.720	0.913	1.091	92.2%	92.2%	93.8%
Score 1	0.066	0.621	−0.325	0.190	0.415	17.2%	37.5%	43.8%
Score 2	0.497	0.614	0.092	0.410	0.848	3.1%	18.8%	25.0%
Score 3	−0.165	0.639	−0.451	−0.097	0.213	3.1%	4.7%	12.5%
UNEMP	−0.079	0.087	−0.105	−0.060	−0.018	20.3%	35.9%	40.6%
F-test on Scores 1–3						12.5%	29.7%	45.3%
Adj. R^2	48.2%	25.9%	29.0%	50.2%	64.9%			
No. of obs.	131.6	45.1	105.3	128.0	168.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

where $E_{j,t}\tau_t$ is respondent j's forecast of the trend rate of inflation at time t. This is the long-horizon forecast CPI inflation rate, which does not depend on h.

To empirically construct the variable $E_{j,t}\tau_t$, we use the combination of SPF 10-year CPI inflation forecasts and Blue Chip Economic Indicators. For the period 1991:Q4 to 2022:Q2 inclusive, we use the individual forecasts of average 10-year CPI inflation (CPI10). Prior to 1991:Q4 this variable was not collected, and for the surveys from 1981:Q3 to 1991:Q3 we use the twice yearly long-term inflation forecasts from Blue Chip Economic Indicators, made available on the SPF website (as Additional-CPIE10.xslx). We lag the data, so that the 1981:Q3 survey uses the 1981:Q2 Blue Chip figure, etc. Thus, the trend inflation rate varies across respondents from 1991:Q4 onwards, but prior to that date it is the same for all respondents.

Table 13 shows the summary of parameter estimates of Model (11) across respondents. Compared to our baseline results, there is a marginal reduction in the percentage of rejections of the t-test of Score 1, and of the F-test of all scores together. However, the percentage of rejections for the t-test of Score 3 is almost doubled, from 7.8% to 15.5%. We conclude that adoption of an inflation gaps formulation does not qualitatively alter our findings.

One might argue that the unemployment rate term should enter as a 'gap', that is, that inflation rate forecasts depend on the difference between the unemployment rate and the natural rate. However, forecasts of the natural rate were only collected by the U.S. SPF from 1996:Q3 onwards, and then only for the third quarters of the year. Clements (2024) uses the estimate of the natural rate available at the time of the survey as the forecast, and finds the results are little affected. In Appendix E, we replace the forecast of the unemployment rate with the first difference of the unemployment rate forecast, with little change to the results.

5. Conclusions

We set out a way of applying FDA to a specific application using survey expectations. We estimated Phillips curve models at an individual level for a sample of respondents to the U.S. SPF over the last 40 years. We estimated the respondents' implicit beliefs in the responsiveness of inflation to the unemployment rate, based on their reported forecasts, while at the same time allowing that forecasters might be influenced by the forecasts of others. We investigated the potential of FDA as a way of allowing for the influence of others' inflation forecasts. We found that for nearly half of the professional forecasters in the sample, there was a role for the earlier inflation forecasts of the other respondents.

We assessed whether the FDA scores were simply capturing the first and second moments of the cross-sectional distribution of prior forecasts. One might suppose that forecasters adjust their forecasts towards the consensus view (i.e. herding), or indeed deliberately move away from the consensus position (anti-herding). However, the FDA scores transpired to be statistically significant for some respondents even when the consensus forecast was explicitly included. In fact, they remained significant for some respondents even when the cross-sectional standard deviation and higher moments of the forecast distribution were included. The outcomes of encompassing tests of models with FDA scores against models with

Table 13
Summary of parameter estimates of Model (11) across respondents.

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.349	0.479	0.072	0.322	0.632	34.5%	43.1%	50.0%
CPI Gap	0.752	0.282	0.576	0.785	0.949	93.1%	94.8%	98.3%
Score 1	0.035	0.494	−0.192	0.110	0.317	27.6%	37.9%	48.3%
Score 2	0.344	0.439	0.097	0.258	0.563	3.4%	17.2%	27.6%
Score 3	0.039	0.431	−0.127	0.064	0.274	3.4%	15.5%	19.0%
FD UNEMP	−0.074	0.071	−0.119	−0.069	−0.025	36.2%	44.8%	56.9%
F-test on Scores 1–3						27.6%	39.7%	50.0%
Adj. R^2	38.7%	24.0%	20.0%	35.1%	48.2%			
No. of obs.	232.9	84.0	161.0	212.0	285.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

cross-sectional moments depended on whether we considered forecasts made during recessions or normal periods, and suggested that the moments and functional factors may capture different influences.

Alongside the empirical application, we provided a discussion of some aspects of FDA in the hope that this will encourage its wider use. We know of only one other study that makes use of FDA with survey expectations data (Meeks and Monti (2023)), and its application in economics has been more limited than in other disciplines (but see the papers referred to in the introduction). FDA has the potential to capture the diversity of beliefs, as in Meeks and Monti (2023), or perhaps to serve as a complement to alternative approaches, such as choosing a selection of moments from the cross-sectional distribution of responses, as in our study of forecaster behavior. Whether it ends up as a complement or viable alternative to the use of cross-sectional moments is largely an empirical question, and we hope that our application will encourage further investigation of its potential.

CRediT authorship contribution statement

Michael P. Clements: Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Shixuan Wang:** Writing – review & editing, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

This is the appendix for the article 'Do professionals' inflation forecasts incorporate the beliefs of others? A functional data approach'. In Appendix A we present the results for a hybrid Phillips curve model. In Appendix B we decompose Score 2 into positive and negative parts. Appendix C checks for horizon fixed effects, and Appendix D checks whether our findings are robust to allowing for a measure of uncertainty concerning the economic environment. In Appendix E we replace the unemployment rate with the first difference of the unemployment rate to proxy the unemployment rate gap, and finally, Appendix F considers the robustness of the results to the treatment of outliers.

Appendix A. The specification of the Phillips curve model

From a theoretical perspective, a Phillips curve model with a forward-looking inflation term is better grounded than one with a backward-looking term, although one could consider a hybrid Phillips curve model which includes both. For example, Clements (2024) used a hybrid Phillips curve model that includes both a forward-looking term ($E_{j,t}\pi_{t+h+1}$) and a backward-looking term ($E_{j,t}\pi_{t+h-1}$). Following this approach, we modify our Model (7) to a hybrid Phillips curve version as:

$$E_{j,t}\pi_{t+h} = \delta_j + \beta_j E_{j,t}\pi_{t+h+1} + \psi_j E_{j,t}\pi_{t+h-1} + \sum_{k=1}^{3} \hat{\xi}_{k,t}\gamma_{j,k} + \alpha_j E_{j,t}u_{t+h} + e_{j,t}. \quad (12)$$

Note that we can only consider $h = 1, 2, 3$ (without $h = 0$) because we do not use $E_{j,t}\pi_{t-1}$ in this study. As a corresponding adjustment, we select the respondents who made 90 or more forecasts (for all $h = 1, 2, 3$), which is still at least 30 forecasts per h on average. By this adjustment, the same 64 respondents are selected as in Section 3.2.

Table 14 presents the summary statistics of the parameter estimates for Model (12) for each respondent. Compared to the baseline model, we find a reduction in the percentage of respondents for whom the scores are individually statistically significant, and a reduction in the percentage of rejections for the three scores jointly.

Table 14
Summary of parameter estimates of Model (12).

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.511	0.767	0.073	0.273	0.734	25.0%	42.2%	54.7%
CPI_forward	0.585	0.250	0.471	0.637	0.726	92.2%	96.9%	96.9%
CPI_backward	0.206	0.122	0.146	0.220	0.272	90.6%	96.9%	96.9%
Score 1	−0.088	0.360	−0.160	−0.021	0.087	14.1%	21.9%	31.3%
Score 2	0.259	0.401	0.014	0.167	0.403	14.1%	18.8%	29.7%
Score 3	−0.042	0.407	−0.161	−0.017	0.097	1.6%	15.6%	17.2%
UNEMP	−0.014	0.046	−0.035	−0.006	0.007	4.7%	23.4%	25.0%
F-test on Scores 1–3						18.8%	26.6%	31.3%
Adj. R^2	64.9%	25.2%	50.8%	66.7%	85.6%			
No. of obs.	182.3	67.2	122.3	166.5	234.5			

Note: The estimates are based on individual systems of three equations (i.e. one equation for $h = 1, 2, 3$), for each respondent. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Nevertheless, we still find that we reject the joint insignificance of the scores for over a quarter of respondents in this more highly parameterized model.

Appendix B. Decomposing score 2 into positive and negative values

Given the interest in tail-end growth risks in the recent literature (see e.g. Adrian et al. (2019) and Adams et al. (2021)), we consider the possibility that forecaster behavior might be affected differently by upside and downside risk. In our setting, this means allowing that forecasters might respond differently to positive versus negative skew in (last period's) cross-section of forecasts. To investigate this possibility, we decompose Score 2 as:

$$\hat{\xi}_{2,t}^{(+)} = \max\left\{0,\ \hat{\xi}_{2,t}\right\},$$
$$\hat{\xi}_{2,t}^{(-)} = \min\left\{0,\ \hat{\xi}_{2,t}\right\},$$

where $\xi_{2,t}^{(+)}$ captures the positive skew, and $\xi_{2,t}^{(-)}$ measures the negative skew. Then we run the following regression:

$$E_{j,t}\pi_{t+h} = \delta_j + \phi_j E_{j,t}\pi_{t+h+1} + \hat{\xi}_{1,t}\gamma_{1,k} + \hat{\xi}_{2,t}^{(+)}\gamma_{2,k}^{(+)} + \hat{\xi}_{2,t}^{(-)}\gamma_{2,k}^{(-)} + \hat{\xi}_{3,t}\gamma_{3,k} + \alpha_j E_{j,t}u_{t+h} + e_{j,t}. \quad (13)$$

Table 15 provides the summary statistics of the parameter estimates for Model (13) across respondents. We find fewer rejections for either $\gamma_{2,k}^{(+)}$ or $\gamma_{2,k}^{(-)}$ compared to the baseline model, suggesting that there is no evidence of a different response to positive and negative values.

Appendix C. Fixed effects of *h*

A simple way to check the assumption of the constancy of parameters over the forecast horizon h, in the main text, is to check for the significance of horizon fixed effects as follows:

$$E_{j,t}\pi_{t+h} = \delta_j + \beta_j E_{j,t}\pi_{t+h+1} + \sum_{k=1}^{3}\hat{\xi}_{k,t}\gamma_{j,k} + \alpha_j E_{j,t}u_{t+h} + \eta_{j,h} + e_{j,t}. \quad (14)$$

where $\eta_{j,h}$ contains the fixed effects h for respondent j.

Table 16 shows the summary of parameter estimates of Model (14) across respondents. The results allowing for fixed effects are very similar to the results for the baseline model without fixed effects. Moreover, the estimated coefficients on the horizon dummies are significant for only 12%–17% of the 64 respondents.

Appendix D. Inclusion of an uncertainty measure capturing economic conditions: VIX

We add the VIX as the additional regressor in our baseline model and check whether our results are still robust. We use the quarterly (end of quarter) VIX data downloaded from Federal Reserve Economic Data (FRED). Note that VIX starts from 1990:Q1, and thus our results with VIX are based on the period from 1990:Q1 to 2022:Q2. Table 17 presents the summary of parameter estimates across respondents of Model (7) with VIX included as the additional regressor. The results with VIX are very similar to the baseline results, suggesting the robustness of our results to the addition of VIX.

Appendix E. Use of the first difference of the unemployment rate forecasts

We replace the level of the unemployment rate forecast with the first difference of the unemployment rate forecast:

$$E_{j,t}\pi_{t+h} = \delta_j + \beta_j E_{j,t}\pi_{t+h+1} + \sum_{k=1}^{3}\hat{\xi}_{k,t}\gamma_{j,k} + \alpha_j \Delta E_{j,t}u_{t+h} + e_{j,t} \quad (15)$$

where $\Delta E_{j,t}u_{t+h} = E_{j,t}u_{t+h} - E_{j,t-1}u_{t+h-1}$.

The result is shown in Table 18. There is a marginal drop in the percentage of rejection in the t-test of Score 1 and the F-test.

Appendix F. Remove outliers at the 5% level

When working with the LQD, it is necessary to remove outliers to prevent values of infinity occurring (as when any $s \in (0, 1)$ gives $f_t(Q_t(s)) = 0$). However, the choice of the level at which to remove outliers is somewhat

Table 15
Summary of parameter estimates of Model (13).

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.751	0.744	0.284	0.748	1.135	43.8%	57.8%	68.8%
CPI	0.825	0.300	0.679	0.879	1.020	95.3%	96.9%	96.9%
Score 1	0.096	0.464	−0.225	0.164	0.364	28.1%	46.9%	54.7%
Score 2 - positive	0.427	0.771	−0.099	0.252	0.730	3.1%	6.3%	14.1%
Score 2 - negative	0.316	0.694	−0.017	0.238	0.616	0.0%	6.3%	9.4%
Score 3	−0.001	0.467	−0.212	0.022	0.211	4.7%	7.8%	10.9%
UNEMP	−0.069	0.069	−0.099	−0.056	−0.015	93.8%	98.4%	100.0%
F-test on Scores 1–3						25.0%	37.5%	48.4%
Adj. R^2	46.7%	25.0%	28.4%	46.6%	64.0%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Table 16
Summary of parameter estimates of Model (14) across respondents.

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.756	0.787	0.285	0.727	1.237	40.6%	64.1%	68.8%
CPI	0.826	0.303	0.683	0.888	1.024	95.3%	96.9%	96.9%
Score 1	0.081	0.467	−0.213	0.153	0.363	25.0%	46.9%	53.1%
Score 2	0.379	0.458	0.028	0.279	0.606	7.8%	21.9%	28.1%
Score 3	0.010	0.450	−0.182	0.000	0.286	3.1%	9.4%	14.1%
UNEMP	−0.066	0.069	−0.098	−0.053	−0.017	34.4%	40.6%	40.6%
FE: $h = 1$	−0.017	0.216	−0.106	0.011	0.114	4.7%	14.1%	21.9%
FE: $h = 2$	−0.016	0.225	−0.093	0.004	0.133	4.7%	17.2%	23.4%
FE: $h = 3$	−0.009	0.228	−0.109	0.002	0.096	6.3%	12.5%	21.9%
F-test on Scores 1–3						25.0%	43.8%	56.3%
Adj. R^2	46.9%	24.8%	28.1%	46.6%	64.0%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$ of (14)), for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Table 17
Summary of parameter estimates of Model (7) with VIX across respondents.

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.817	0.839	0.304	0.864	1.319	45.8%	61.0%	67.8%
CPI	0.814	0.314	0.647	0.849	1.034	94.9%	96.6%	96.6%
VIX	−0.004	0.012	−0.011	−0.003	0.002	11.9%	22.0%	28.8%
Score 1	0.096	0.476	−0.201	0.165	0.396	27.1%	45.8%	52.5%
Score 2	0.349	0.481	0.009	0.238	0.573	5.1%	16.9%	28.8%
Score 3	−0.037	0.508	−0.240	−0.013	0.177	3.4%	6.8%	10.2%
UNEMP	−0.062	0.069	−0.098	−0.046	−0.009	32.2%	39.0%	47.5%
F-test on Scores 1–3						22.0%	40.7%	59.3%
Adj. R^2	44.8%	24.1%	28.5%	44.3%	59.9%			
No. of obs.	241.6	88.8	172.0	223.0	300.5			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent j. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

arbitrary. In this section we check that the results are not overly sensitive to the choice made. In the baseline analysis, outliers at the 1% level in each quarter t for a given h are removed. This is implemented by excluding observations below the first percentile value and the above-99% value in each quarter. We repeat the analysis with outliers removed at the 5% level. The summary of parameter estimates of the baseline model is presented in Table 19. We find that the mean of the parameter estimates is similar to the base case. The F-test rejection

Table 18
Summary of parameter estimates of Model (15) across respondents.

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.334	0.605	−0.089	0.331	0.703	45.3%	60.9%	68.8%
CPI	0.833	0.298	0.695	0.884	1.016	95.3%	96.9%	98.4%
Score 1	−0.060	0.410	−0.206	0.024	0.170	23.4%	34.4%	39.1%
Score 2	0.390	0.465	0.044	0.314	0.550	12.5%	18.8%	29.7%
Score 3	0.011	0.470	−0.165	−0.011	0.278	3.1%	9.4%	12.5%
FD UNEMP	−0.082	0.143	−0.169	−0.062	0.000	29.7%	45.3%	45.3%
F-test on Scores 1–3						20.3%	35.9%	45.3%
Adj. R^2	46.8%	24.7%	27.5%	46.8%	60.9%			
No. of obs.	241.6	89.6	161.5	220.5	304.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent j. FD UNEMP is the first difference of the unemployment rate forecast. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels.

Table 19
Summary of parameter estimates of Model (7) with outliers removed at the 5% level.

	mean	s.d.	l.q.	median	u.q.	rej. 1%	rej. 5%	rej. 10%
Intercept	0.735	0.750	0.325	0.785	1.066	48.4%	60.9%	75.0%
CPI	0.826	0.298	0.675	0.867	1.016	95.3%	98.4%	98.4%
Score 1	0.087	0.463	−0.228	0.164	0.358	28.1%	39.1%	50.0%
Score 2	0.472	0.532	0.083	0.348	0.679	17.2%	35.9%	39.1%
Score 3	0.078	0.487	−0.149	0.012	0.381	1.6%	10.9%	14.1%
UNEMP	−0.066	0.066	−0.095	−0.052	−0.017	31.3%	39.1%	43.8%
F-test on Scores 1–3						35.9%	46.9%	57.8%
Adj. R^2	47.0%	24.7%	28.9%	46.5%	63.6%			
No. of obs.	243.5	89.6	162.8	223.0	312.8			

Note: The estimates are based on individual systems of four equations (i.e. one equation for $h = 0, 1, 2, 3$), for each respondent. For each parameter, we present the summary statistics of the cross-sectional distribution over j, including the mean, standard deviation (s.d.), lower quartile (l.q.), median, and upper quartile (u.q.). In the last three columns, we report the proportion of the 64 regressions for which we reject the null hypothesis of the parameter equal to zero at 1%, 5%, and 10% levels. Outliers at the 5% level in each quarter t for a given h are removed.

percentage (at the 5% significance level) of 47% is very close to the base case. The rejection percentage for the individual scores is slightly lower for Score 1 but a little higher for Scores 2 and 3. We conclude that the findings are not overly sensitive to the treatment of outliers.

Data and code availability

The reproducibility package, including data and code, for reproducing the results in the paper is available here: https://github.com/wsx423104361/Reproducibility_FDA_SPF.

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